Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3471-WE01

Title: (BEng/MEng)

Materials 3

Time Allowed:	2 hours
Additional Material provided:	Yes, formula sheet included as part of exam paper
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Section A
Question 1

a) What makes plain carbon steel harder than pure iron? Do dislocations control the strength of a silicate glass? Explain.

[10%]

b) A construction site wants to use low-carbon steel in a condition where the temperature is 80°C in a corrosive environment. An engineer has suggested the following materials as potential coatings: Poly(ethylene adipate), Alcad, Graphite and 1100 aluminum. Are these suggestions feasible? Discuss the suitability of each coating in terms of corrosion.

[25%]

c) An engineer has analyzed the materials and found that its ASTM grain size number is 26 and all grains are a square shape and the same size under the microscope. How much is the yield stress of the materials? Yield stress (strength) of the material against grain size is illustrated in Figure Q1c.

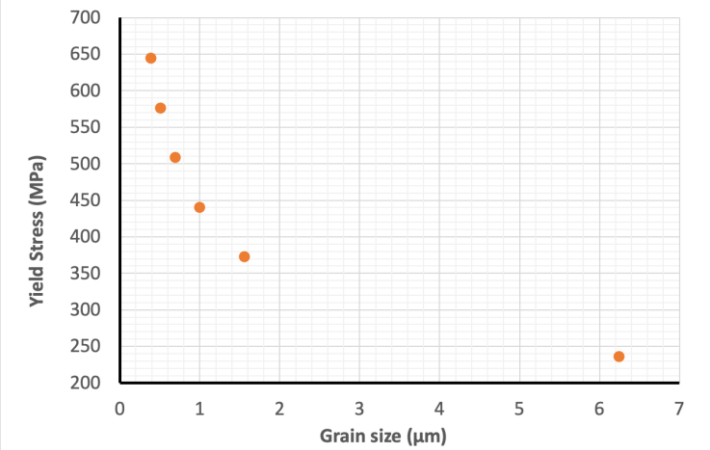


Figure Q1c: Yield stress vs the grain size of a matreials used in Q1C.

[40%]

d) In view of the phase diagram shown below (cf. Figure Q1d), can you identify all phases and their composition at 500 and 1000°C, respectively.

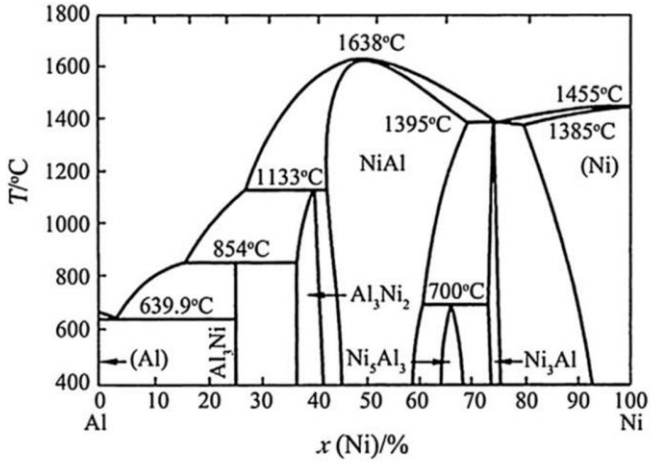


Figure Q1d: Phase digram of Al-Ni.

[25%]

continued

Section A
Question 2

a) Oxidation of most ceramics is not considered to be a problem. Explain.

[10%]

b) Plastic deformation due to dislocation motion is important in metals; however, this is not a very important consideration for the properties of polymers. Explain.

[10%]

c) By considering the phase diagram shown in Figure Q2, how many grams of Na_2O can be added to 350 g of SiO_2 to keep the melting point of the alloy to the lowest level? Molecular weights of Na, Si and O are 23.0, 28.1 and 16 g/mol, respectively.

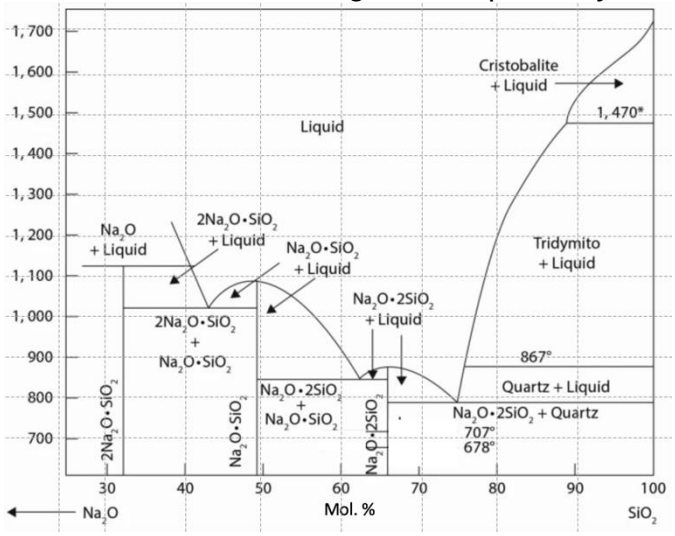


Figure Q2c: Phase diagram of the Na_2O-SiO_2 system

[50%]

d) For a polymer operating at constant strain, we apply a stress of 10MPa; after 180 days, the stress drops to 7 MPa. An application requires a part made from the same polymer to maintain 8.5 MPa after 40 weeks. How much stress should be applied initially to the polymer in this scenario?

[30%]

continued

Section B
Questions 3

a) Among the cases presented below, which one(s) is/are best explained by LCA (Life Cycle Assessment), and which is/are not? Provide a detailed explanation.

i. Should we use steel or glass fibre composite for our cars' bodies considering their environmental impact?
ii. What is more environmentally friendly, washing dishes by hand or using a dishwasher?
iii. Are polar bears increasingly becoming hermaphrodites as a result of global emissions of endocrine disruptors?
iv. Would increasing taxes on old diesel cars reduce particle emissions and thereby reduce hospital spending on treating lung diseases?

[100%]

Section B
Questions 4

You are considering material choice for the ladder chassis (highlighted in figure Q4) of a new heavy goods vehicle chassis, seeking to reduce the CO_2 production associated with the vehicle.

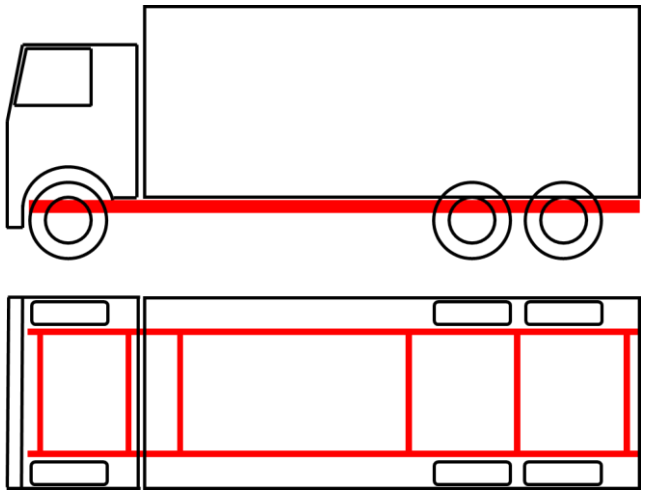


Figure Q4 – HGV with chassis structural elements highlighted in red.

a) A colleague sources a table of properties of candidate materials (Table QT4) and proposes that the best choice would be the material with the lowest value in the "carbon footprint" column. Briefly explain why your colleague's approach is flawed.

[25%]

b) Identify the appropriate material index from the point of view of the CO_2 associated with the production of the raw material. You do not need to derive it from first principles, but you should justify your methodology.

Rank the candidate materials based on your metric. **[25%]**

Making the vehicle lighter could provide an energy and CO_2 saving over the life of the vehicle due to reduced rolling resistance and energy to accelerate the vehicle. For a diesel truck on long haul service the additional powertrain CO_2 output per km for each additional kg of vehicle mass can be taken to be 0.035 $\frac{\text{grams}CO_{2e}}{\text{kg km}}$. The design life of the truck is 1,200,000 km.

c) What is the appropriate combination of material properties to be maximised to minimise the combined CO_{2e} considering the combination of production of the raw materials and the CO_{2e} emitted by the vehicle over its lifetime?

Rank the candidate materials based on this revised metric. **[25%]**

d) Your company moves forward with the development of a new truck with a lighter chassis to provide a lifetime CO_2 saving. However, if truck operators decide instead to take the opportunity of the lighter chassis to increase the payload that they can carry on each journey so that the loaded truck will end up being just as heavy as its predecessor. Does this undermine the benefit that you have achieved? Explain your rationale.

[25%]

continued

Material	E Young's Modulus (GPa)	σ_f Failure Stress (MPa)	ρ Density (kg/m^3)	C_F Carbon Footprint ($kgCO_{2e}/kg$)
Aluminium Alloy 6083T6	70.0	240	2700	12.5
Magnesium Alloy	44.5	220	1850	35.5
Steel (low carbon)	210	355	7850	1.9
Titanium Alloy 6AL4v	114	880	4430	46.5

Table QT4 – Material Properties and Indices

continued

Section B
Questions 5

(a) Metalex is an Engineering firm manufacturing bespoke metal products for customers. They have received an order to produce 100 cast iron door knockers in 10 days. Upon testing the product, they find that the door knockers develop internal stresses and become hard and brittle. In your opinion, what heat treatment process should they use? Explain your reasoning.

[40%]

(b) Powder metallurgy is a metal processing technology in which parts are produced from metallic powders. Describe five applications that power metallurgy techniques can be used for.

[20%]

(c) A prototype of a tube with a square cross section is to be fabricated using fused deposition modelling. The outside dimension of the square = 110 mm and the inside dimension = 106 mm (wall thickness = 2 mm, except at corners). The height of the tube (z-direction) = 60 mm, the layer thickness = 0.25mm, and the width of the extrudate deposited on the surface of the part = 1.2mm. The extruder workhead moves in the x-y plane at a speed of 250mm/s. A delay of 15s is experienced between each layer to reposition the workhead. Compute an estimate for the time required to build the part.

[40%]

continued

Formula sheet

	Metal	Electrode Potential E^0 (Volts)
Anodic	$Li^+ + e^- \rightarrow Li$	–3.05
	$Mg^{2+} + 2e^- \rightarrow Mg$	–2.37
	$Al^{3+} + 3e^- \rightarrow Al$	–1.66
	$Ti^{2+} + 2e^- \rightarrow Ti$	–1.63
	$Mn^{2+} + 2e^- \rightarrow Mn$	–1.63
	$Zn^{2+} + 2e^- \rightarrow Zn$	–0.76
	$Cr^{3+} + 3e^- \rightarrow Cr$	–0.74
	$Fe^{2+} + 2e^- \rightarrow Fe$	–0.44
	$Ni^{2+} + 2e^- \rightarrow Ni$	–0.25
	$Sn^{2+} + 2e^- \rightarrow Sn$	–0.14
	$Pb^{2+} + 2e^- \rightarrow Pb$	–0.13
	$2H^+ + 2e^- \rightarrow H_2$	0.00 — (defined)
	$Cu^{2+} + 2e^- \rightarrow Cu$	+0.34
	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	+0.40
	$Ag^+ + e^- \rightarrow Ag$	+0.80
	$Pt^{4+} + 4e^- \rightarrow Pt$	+1.20
	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	+1.23
Cathodic	$Au^{3+} + 3e^- \rightarrow Au$	+1.50

The standard reduction potentials for selected elements and reactions

Active, Anodic End	Magnesium and Mg alloys
	Zinc
	Galvanized steel
	5052 aluminum
	3003 aluminum
	1100 aluminum
	Alclad
	Cadmium
	2024 aluminum
	Low-carbon steel
	Cast iron
	50% Pb-50% Sn solder
	316 stainless steel (active)
	Lead
	Tin
	Cu-40% Zn brass
	Nickel-based alloys (active)
	Copper
	Cu-30% Ni alloy
	Nickel-based alloys (passive)
	Stainless steels (passive)
	Silver
	Titanium
	Graphite
	Gold
	Platinum
	Noble, Cathodic End

The galvanic series in seawater

Bragg's Law:

$$sin\ \theta = \frac{\lambda}{2d_{hkl}}$$

continued

Concentration of vacancies:

$$n_v = n\exp(-Q_v/RT)$$

ASTM grain size number:

$$N = 2^{n-1}$$

Percent cold work:

$$Percent\ of\ cold\ work = \left[\frac{A_0 - A_f}{A_0}\right] \times 100$$

Hall-Petch Equation:

$$\sigma_y = \sigma_0 + kd^{-1/2}$$

Stress relaxation:

$$\sigma = \sigma_0\exp(-t/\lambda)$$

Relaxation time:

$$\lambda = \lambda_0\exp(Q/RT)$$

Formula sheet - Material Processing

Screen mesh for sorting particle sizes

$$PS = \frac{1}{MC} - t_w$$

Where PS = particle size, MC = mesh count, openings per linear mm and t_w = wire thickness of screen mesh.

Surface area (sphere)

$$A = \pi D^2$$

Where D = diameter of the spherical particle

Surface volume (sphere)

continued

$$V = \frac{\pi D^3}{6}$$

Where D = diameter of the spherical particle

Area-to-volume ratio for a sphere

$$\frac{A}{V} = \frac{6}{D}$$

Where A = surface area, V = surface volume, D = diameter of the spherical particle

Area-to-volume ratio for any particle shape – spherical or non-spherical

$$\frac{A}{V} = \frac{K_S}{D} \text{ or } K_S = \frac{AD}{V}$$

Where K_S = shape factor; D = the diameter of a sphere of equivalent volume as the nonspherical particle.

continued

Power density

$$PD = \frac{P}{A}$$

where PD = power density, W/mm^2; P = power entering the surface, W; and A= surface area over which the energy is entering, mm^2.

Heat required to melt a given volume of metal

$$U_m = KT_m^2$$

where U_m = the unit energy for melting, J/mm^3; T_m = melting point of the metal on an absolute temperature scale, °K(°R); and K = constant whose value is 3.33 x 10^{-6} JK^{-2}mm^{-3}, when the Kelvin scale is used (and K = 1.467 x 10^{-5} JR^{-2}mm^{-3} for the Rankine temperature scale).

Heat energy available for welding:

$$H_w = f_1 f_2 H$$

where H_w = net heat available for welding, J, f_1 = heat transfer factor, f_2 = the melting factor, and H = the total heat generated by the welding process, J.

Energy balance equation

$$H_w = U_m V$$

Where H_w = net heat energy used by the welding operation, J; U_m = unit energy required to melt the metal, J/mm^3; and V = the volume of metal melted, mm^3.

Rate balance equation

$$R_{Hw} = U_m R_{WV}$$

Where R_{Hw} = rate of heat energy delivered to the operation for welding, J/s = W; and R_{WV} = volume rate of metal welded, mm^3/s.

$$R_{Hw} = f_1 f_2 R_H = U_m A_w v$$

here f_1 and f_2 are the heat transfer and melting factors; R_H = rate of input energy generated by the welding power source, W; A_w = weld cross-sectional area, mm^2; and v = the travel velocity of the welding operation, mm/s.

continued

Time to complete a single layer in rapid prototyping

$$T_i = \frac{A_i}{vD} + T_r$$

Where T_i= time to complete layer i; A_i = area of layer i; v = average scanning speed of the laser beam at the surface; D = diameter of the laser beam at the surface, and T_r = repositioning time between layers.

Build cycle time (stereolithography)

$$T_c = \sum_{i=1}^{nl} T_i$$

Where T_c = the STL build cycle time; and nl = the number of layers used to approximate the part.

END